A Blockchain-Based Approach to Carbon Footprint Monitoring in Early-Stage Olive Oil Production

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Why blockchain?

- Carbon claims should travel with the product, not sit in a separate LCA report.
- In the first three phases, emissions can still be tied to a specific olive lot and a responsible actor.
- Each phase becomes one signed, hash-linked block that certifiers and buyers can audit.

Case study & method

Location	Western Achaia, Greece
Orchard	Koroneiki · ~80 yr · 4.8 ha
Seasons	2022 · 2023 · 2024
Method	LCA per ISO 14067 (gross)
Scope	Orchard → transport → mill

Three-block traceability architecture

BLOCK 1
Cultivation & harvesting

Actor: Olive grower

- Orchard & lot ID, crop year
- Soil prep, fertilisation, plant protection
- Pruning, harvesting, input quantities
- Farm-stage emissions

81.5% of total emissions

BLOCK 2
Transport to olive mill

Actor: Grower/transporter

- Linked lot ID
- Loading & delivery event
- Route or distance, quantity
- Transport emissions

1.5% of total emissions

BLOCK 3
Processing at olive mill

Actor: Mill operator

- Mill ID, lot reception
- Electricity & water use
- Extraction operations
- Oil output & processing emissions

17.0% of total emissions

✓ Each block is countersigned by an independent validator — agronomist or inspection body for Block 1, consignor + consignee for Block 2, metered readings for Block 3 — so the ledger stores values that are both immutable and trustworthy.

Olive lot → **Oil batch** → **Bottle on shelf**

One lot ID travels to the label: scan the bottle, read the signed chain.

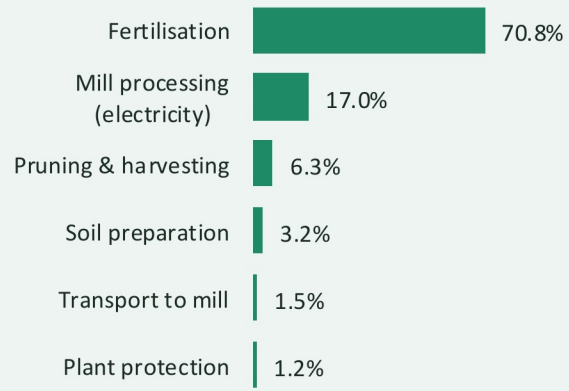
Carbon footprint of the studied EVOO

1.993 kg CO_2eq / L

Gross emissions, orchard to mill. Orchard carbon sequestration not credited.

Emission hotspots

share of total emissions, %



1. **81.5% at the orchard** — Fertilisation alone is 70.8% — Block 1 holds the lever.
2. **Permissioned, not proof-of-work** — Known validating nodes; ledger energy is negligible.
3. **Next: packaging block** — Bottle-level IDs, orchard sequestration term, full net balance.

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